Indian Institute of Technology Madras

BS Degree Programme

Data Visualization Design Project (May 2026)

A Visual Study of E-Commerce Orders, Delivery & Customer Satisfaction

Balancing Marketplace Growth with Customer Experience in Brazilian E-Commerce

Team Member Name	Student ID
Sanjoli (Team Lead)	23F1002254
Jay Rajesh Mestry	22F3000169
Joyce Jemima S	21F1006131
Kishar Kr Nath	21F3000476
P. Vishal Reddy	21F1001676

Submission Date: September 11, 2026

Contents

List of Figures

List of Tables

1 Executive Summary & Problem Statement

1.1 Executive Summary

This report analyzes two years of operational data (late 2016 to mid-2018) from the Brazilian e-commerce platform Olist, covering roughly 100,000 orders, 112,000 items, and 8,000 seller leads [1]. During this period, order volume grew twentyfold while maintaining an overall delivery completion rate of 97%.

Despite this rapid growth, several operational issues affect customer retention. Delivery timing is the primary source of customer dissatisfaction. Most orders arrive ahead of schedule, but the 8% of orders delivered after the estimated date account for most of the platform's 1-star reviews. Geographic concentration also creates logistics difficulties. Customers in São Paulo generate roughly 42% of all orders, while shipments sent to distant states face long transit times, high freight rates, and frequent complaints.

Internal processes also create friction. When an order contains items from multiple sellers, packages arrive separately and average review ratings drop. In addition, merchant acquisition has high drop-off rates, as more than half of newly onboarded sellers never complete a sale.

1.2 Problem Statement: Growth vs. Customer Experience

The marketplace management team faces a difficult trade-off between expanding platform scale and maintaining service quality.

Bringing in more sellers increases product selection and total sales volume, but it also increases the risk of slow dispatch times, fragmented deliveries, and negative reviews. Conversely, setting strict requirements on seller performance protects customer trust, but can slow down merchant growth and reduce catalog variety.

Until now, platform managers have balanced these priorities without clear metrics. This report evaluates the complete order lifecycle, from seller sign-up to customer delivery, to provide clear visual evidence for these decisions.

1.3 Key Questions Addressed

This analysis focuses on five main questions:

1. **Delivery Delay Impact:** How significantly do late deliveries and missed promise dates reduce customer review scores?
2. **Multi-Seller Orders:** How does splitting a single order across different merchants affect delivery performance and customer ratings?
3. **Regional Logistics Bottlenecks:** Which interstate routes and destination states experience the most severe transit delays?
4. **Seller Onboarding Quality:** Which acquisition channels bring active, high-volume merchants rather than inactive accounts?
5. **Payment Methods and Installments:** How does installment-based payment affect order values and fulfillment timelines?

2 Data Preparation & Cleaning

Before running our exploratory and explanatory analysis, we cleaned, deduplicated, and transformed both the e-commerce transaction data and the marketing funnel records [1]. This ensured that operational timelines, customer review ratings, and geographic distances were consistent across all tables.

2.1 Missing Value Treatment and Deduplication

2.1.1 Geolocation Centroids

The raw geolocation table contained over one million coordinate rows, with dozens of duplicate entries for individual postal codes. We grouped the coordinates by postal code prefix (`geolocation_zip_code_prefix`) and calculated median latitude and longitude values. This reduced the table to 19,015 unique coordinate centroids. Coordinate points that fell outside Brazil's geographic boundaries were removed as data entry errors.

2.1.2 Review Text Fields

Roughly 88% of review titles and 58% of review comments were left blank by customers. However, all 99,224 records contained a valid numeric rating between 1 and 5 stars. We retained all rows to preserve the complete satisfaction sample, treating missing text fields as unwritten feedback rather than dropping or imputing them.

2.1.3 Product Dimensions and Packaging

For 610 products lacking physical specifications, we imputed missing weight, length, height, and width values using the median measurements of their specific product category. This preserved item counts and allowed us to compute package volume and freight metrics accurately.

2.1.4 Category Translation

Product category names were translated from Portuguese to English using the supplementary translation table. A small number of unmapped categories (such as `pc_gamer`) were mapped manually so that sales were not lost during category-level grouping.

2.2 Date Parsing and Timeframe Selection

2.2.1 Timestamp Conversion

All date and timestamp columns across the orders, customer reviews, and marketing tables were converted into native date-time format to enable chronological sorting and arithmetic.

2.2.2 Analysis Window

We restricted the primary analysis to the period between January 2017 and August 2018. The platform was just launching in late 2016 with very low, irregular transaction volume, and September 2018 contains only partial tracking data. Isolating these twenty months provides a consistent view of steady-state marketplace operations.

2.2.3 Funnel Velocity Corrections

In the marketing funnel data, one closed deal recorded a closing timestamp slightly before the initial contact timestamp, resulting in a negative duration. This logging error was adjusted to zero days so it would not distort average sales cycle calculations.

2.3 Feature Engineering

To evaluate shipping reliability, merchant performance, and marketing conversion, we derived several key operational variables. Table 1 lists each engineered feature, its underlying data source, and its analytical purpose.

Table 1: Key Engineered Features Across the Fulfillment and Marketing Funnel

Feature Name	**Source Table**	**Calculation and Purpose**
`delivery_delay_days`	`orders`	Actual delivery date minus estimated delivery date. Positive values represent late deliveries.
`is_late`	`orders`	Binary flag marked true for any shipment delivered after the estimated promise date (delay > 0).
`distance_km`	`geolocation`	Great-circle distance between customer and seller zip code centroids, computed using the Haversine equation.
`product_volume_cm3`	`products`	Product length $\times$ width $\times$ height in cubic centimeters, used to evaluate parcel bulk and freight fees.
`days_to_close`	`closed_deals`	Calendar days between initial marketing qualification and deal closing, measuring lead-to-seller conversion time.

3 Exploratory Data Analysis (EDA)

Our exploratory data analysis synthesizes geographic demographics, fulfillment operations, financial behaviors, and customer satisfaction metrics to map the end-to-end customer journey [1].

3.1 Data Architecture & Data Quality Audit

The foundation of this analysis rests on a relational schema connecting customers, orders, payments, products, and sellers. A thorough data quality audit was conducted prior to deeper modeling to verify structural consistency across all metrics. Table 2 details the record counts, primary keys, identified data anomalies, and their respective resolutions.

Table 2: Comprehensive Data Architecture and Quality Audit across Relational Tables

Table	Records	Primary Key	Data Quality Issues	Resolution / Handling
`customers`	99,441	`customer_id`	No missing values across all columns	Direct 1:1 match to order transactions (96,096 unique individuals).
`geolocation`	1,000,163	`zip_code_prefix`	261,831 exact duplicate rows; 42 coordinates placed outside Brazil	Removed spatial outliers; calculated median coordinates per zip prefix down to 19,015 centroids.
`orders`	99,441	`order_id`	Delivery timestamps null for 2,965 non-delivered orders	Isolated 96,478 successfully delivered orders for fulfillment analysis.
`order_items`	112,650	`order_id, item_id`	13,984 multi-item lines; no nulls	Aggregated item prices and freight values to order-level sums.
`payments`	103,886	`order_id, payment_seq`	3 rows marked as `not_defined`	Removed undefined payment rows; summed transaction amounts per order.
`reviews`	99,224	`review_id`	88.3% missing titles; 58.7% missing text comments	Deduplicated surveys by latest submission timestamp; retained null text as silent ratings.
`products`	32,951	`product_id`	610 missing dimension rows; 4 items with 0g weight	Imputed missing measurements using category-level medians; mapped Portuguese-to-English labels.
`sellers`	3,095	`seller_id`	1,027 zip prefixes under 5 digits	Zero-padded postal code strings to standardized 5-digit format.

3.2 Regional Reach and Geographic Concentration

The marketplace exhibits strong geographic centralization on both sides of the platform:

- **Buyer Concentration in the Southeast:** Three Southeastern states generate two-thirds of total demand. São Paulo represents 41.9% of all orders, followed by Rio de Janeiro (12.9%) and Minas Gerais (11.6%).
- **Seller Concentration:** Merchants are even more heavily concentrated in São Paulo, which hosts 59.8% of all registered sellers. Paraná (11.4%) and Minas Gerais (7.9%) are the next largest seller hubs.
- **Customer Retention Limits:** Repeat purchases are uncommon. Only 3% of buyers (fewer than 3,000 customers) ever placed a second order during the twenty-month window. São

Paulo accounts for over 2,500 of these repeat buyers, while second-place Rio de Janeiro records fewer than 1,000.

3.3 Fulfillment Timelines and Regional Logistics Differences

Overall, 97% of orders reach delivery completion, with cancellations and unavailable items remaining below 1% each. However, transit performance varies considerably across regions.

3.3.1 Turnaround Stages

The typical order lifecycle averages 12.6 calendar days from purchase to delivery:

1. **Payment Settlement:** Averages less than 0.5 days.
2. **Seller Handling:** Merchants take an average of 2.8 days to package items and transfer custody to courier services.
3. **Road Transit:** Courier transit averages 9.3 days on the road.

3.3.2 Delivery Estimates vs. Reality

Platform estimation algorithms add an average buffer of 11 days beyond typical transit times. As a result, the majority of orders arrive ahead of their estimated delivery dates. However, 7,828 orders (7.9%) arrived after the promised date.

3.3.3 Regional Freight Costs and Transit Delays

Distance creates substantial freight overhead for outlying states:

- **Freight Burden:** In the North and Northeast, shipping fees account for 22% to 23% of total order value, compared to 15% in the Southeast.
- **Transit Latency:** Shipments routed from São Paulo to northern states such as Pará or Maranhão require 16 to 17 days on average.
- **Local vs. Interstate Ratings:** Orders shipped locally within the same city arrive in under 2 days, receiving an average rating of 4.26 stars and an 11.0% detractor rate (1-star or 2-star reviews). For interstate deliveries, transit times rise to 9.2 days, dropping average review scores to 4.02 stars and increasing detractors to 16.2%.

3.4 Payment Methods and Installment Financing

Customer payment choices influence settlement speed, delivery punctuality, and basket value:

- **Payment Distribution:** Credit cards account for 73.9% of transactions and over 75% of platform revenue, with an average ticket of R$ 163. Boleto bancário (bank payment slip) handles 19.0% of transactions with an average spend of R$ 145. Vouchers (5.4%) and debit cards (1.5%) account for the remainder.
- **Boleto Settlement Lag:** Bank slips take an average of 29 hours to confirm payment, compared to 16 minutes for credit cards. Because sellers wait for payment confirmation before packing, boleto orders have higher late-delivery rates (8.9% late vs. 8.0% for credit cards).
- **Installment Spending Expansion:** While 68% of credit card shoppers pay in a single installment, installment financing significantly increases average order value. A single charge averages R$ 90 to R$ 120, five installments average R$ 220, and ten installments reach R$ 450 to R$ 500.

3.5 Root Causes of Customer Dissatisfaction

Analyzing customer feedback patterns reveals specific operational triggers for poor ratings. Table 3 summarizes the impact of delivery punctuality, survey timing, and cart composition on customer ratings.

Table 3: Customer Review Ratings and Detractor Rates by Operational Condition

Operational Condition	Operational Context	Average Rating	Detractor Rate
On-Time / Early Deliveries	Delivered on or before estimated date	4.29 stars	9.2%
Late Deliveries	Delivered after promised date (> 0 days delay)	2.57 stars	54.1%
Surveys Sent Pre-Delivery	Review requested before package reached customer	2.77 stars	51.4%
Surveys Sent Post-Delivery	Review requested after confirmed delivery	4.28 stars	9.1%
Single-Seller Orders	Complete basket fulfilled by one merchant	4.17 stars	12.3%
Multi-Seller Split Orders	Cart items split across multiple independent sellers	2.86 stars	47.2%

Several key operational takeaways emerge from these comparisons:

1. **Sharp Drop from Missed Deadlines:** When an order misses its estimated arrival date, satisfaction drops from 4.29 stars to 2.57 stars. For delays exceeding one week, average ratings fall below 1.8 stars.

2. **Premature Review Surveys:** In 8.5% of completed orders (over 8,100 cases), the platform sent out satisfaction surveys before the carrier confirmed delivery. Asking customers to evaluate an unreceived parcel reduces review ratings by roughly 1.5 stars.

3. **Review Text Grievances:** Customers who submit written comments are four times more likely to be dissatisfied than those leaving star ratings alone. Textual review parsing indicates that over 74% of complaints cite delivery issues (delays, missing parcels, or incorrect tracking), while fewer than 9% cite item defects.

4. **Seller Dispatch Concentration:** While 9% of items miss seller dispatch deadlines, 5% of the lowest-performing merchants account for roughly two-thirds of all dispatch breaches.

5. **Linear Freight Stacking on Multi-Item Orders:** Charging full individual freight for duplicate items added roughly R$ 325,000 in shipping fees. Discounting freight by 50% on additional units of the same item would return R$ 96,000 to customers and encourage larger basket sizes without compromising baseline shipping margins.

4 Explanatory Analysis & Findings

This section presents the core empirical findings of the study, linking operational milestones, regional logistics disparities, payment mechanisms, and merchant fulfillment behaviors directly to customer review scores and platform retention [1].

4.1 Data Audit and Structural Validation

The transactional dataset tracks 99,441 unique orders placed by 96,096 distinct individuals (`customer_unique_id`). Across these orders, the catalog layer records 112,650 order items across 98,666 order lines, alongside 103,886 payment transactions.

4.1.1 Catalog Dimensions and Skew

Attribute inspection indicates that while physical package dimensions (length, height, and width) stay clustered between 10 and 30 cm, item weight displays extreme positive skew, with values reaching upward of 40,000 g (Figure 1). Exactly 610 products shared identical missing metadata across category names and physical attributes, and 4 items with zero recorded weight were imputed using category medians.

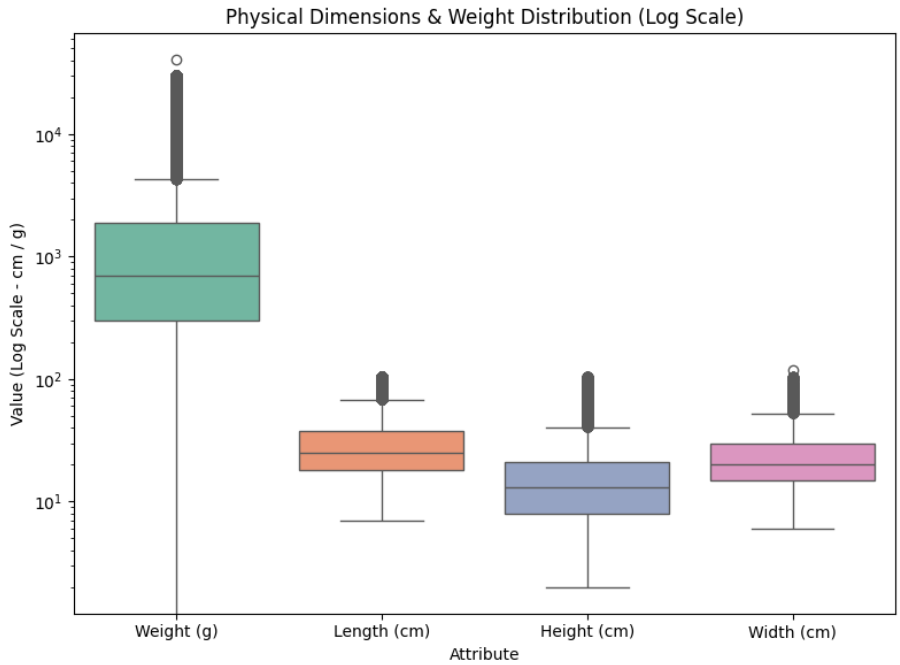

Figure 1: Distribution of Product Measurements: Compact Linear Dimensions vs. Positively Skewed Item Weights.

4.1.2 Financial Reconciliation

Comparing the sum of recorded payment values against item prices plus freight charges confirms an exact match ($|\Delta| < 0.01$) for 99.61% of transactions (98,284 orders). Only 0.25% of records displayed billing differences greater than R$ 1.00, demonstrating strong financial ledger consistency.

4.2 Geographic Concentration and Retention Bottlenecks

Marketplace transaction flows reveal heavy geographic centralization. Both customer purchasing volume and merchant inventory are concentrated in the Southeast, creating fulfillment friction across interstate delivery corridors.

Table 4: Geographic Distribution of Customers, Sellers, and Repeat Buyers

Metric / Dimension	**São Paulo (SP)**	**Rio de Janeiro (RJ)**	**Minas Gerais (MG)**	**Rest of Brazil**
Customer Share	41.98% (41,746)	12.92% (12,852)	11.70% (11,635)	33.40% (24 states)
Merchant Share	59.74% (1,849)	5.53% (171)	7.88% (244)	26.85% (PR: 11.28%)
Repeat Buyers ($\geq$ 2 Orders)	>2,500 buyers	<1,000 buyers	$\approx$730 buyers	Severe retention drop

As summarized in Table 4, São Paulo, Rio de Janeiro, and Minas Gerais together generate 66.60% of all platform purchases. At the city level, the municipal area of São Paulo accounts for the largest share of orders, followed by Rio de Janeiro and Belo Horizonte.

Customer retention remains low across all regions (Figure 2). Only 3.12% of buyers (2,997 individuals) ever completed a second purchase, leaving 96.88% (93,099 individuals) as single-order customers. Repeat purchases drop off steeply outside São Paulo, indicating that prolonged delivery timelines in other states hurt platform loyalty.

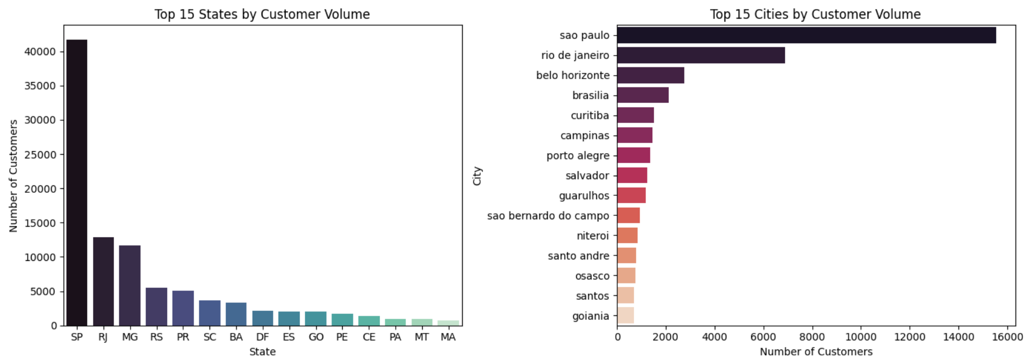

Figure 2: Geographic Concentration of Supply and Demand and the Customer Retention Falloff.

4.3 Fulfillment Timelines, Freight Friction, and Interstate Latency

The imbalance between merchant locations and dispersed customer destinations creates large regional disparities in delivery timelines and shipping expenses.

4.3.1 Fulfillment Milestones

Tracking packages across operational checkpoints indicates that order fulfillment takes an average of 12.56 days (median: 10.22 days):

- **Payment Approval:** Averages 0.43 days (median: 0.01 days, roughly 15 minutes).
- **Merchant Dispatch:** Averages 2.80 days (median: 1.82 days) for sellers to package and hand parcels to carriers.
- **Courier Road Transit:** Averages 9.33 days (median: 7.10 days) on the highway network.

While platform algorithms add an average safety margin of 11.2 days to promised delivery dates, 8.11% of delivered shipments (7,826 orders) still arrived past their deadline.

4.3.2 Regional Freight Disparities and Local Advantages

Shipping fees represent 15.1% of item revenue in the Southeast, but rise to 21.7% in the Northeast and 22.7% in the North. Interstate transit from São Paulo to northern states requires extended delivery times: shipments to Pará (PA) require a median of 17.0 days, and shipments to Maranhão (MA) average 16.4 days.

Conversely, proximity substantially improves customer experience (Figure 3):

- **Same-City Deliveries:** 1.78 days median transit, 4.26 average rating, 11.15% detractor share.
- **Same-State Deliveries:** 4.01 days median transit, 4.19 average rating, 12.36% detractor share.
- **Interstate Deliveries:** 9.08 days median transit, 4.02 average rating, 16.24% detractor share.

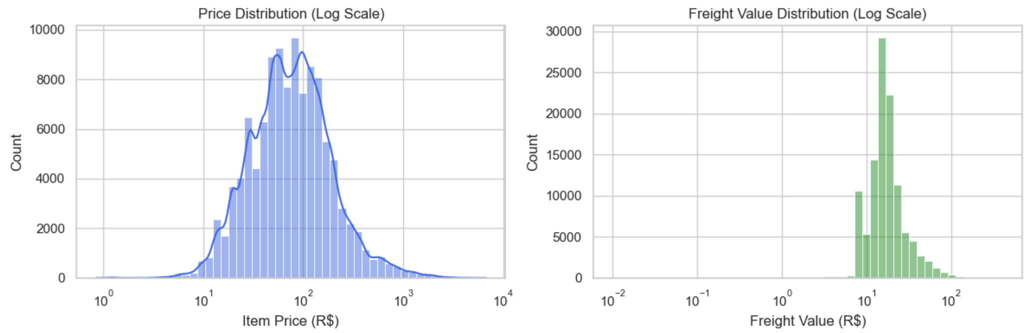

Figure 3: Fulfillment Turnaround Milestones and Regional Delivery Latency Comparison.

4.4 Payment Tender Dynamics and Installment Economics

Customer payment method choices significantly influence transaction sizes, settlement delays, and fulfillment punctuality.

4.4.1 Payment Methods and Settlement Delay

Credit cards represent 73.9% of transactions with an average spend of R$ 163.32, followed by boleto bancário (19.1% share, R$ 145.03 average spend), vouchers (5.6% share, R$ 65.70 average spend), and debit cards (1.5% share, R$ 142.57 average spend), as shown in Figure 4.

Boleto clearance creates an operational delay. Bank slips require a median clearance time of 29.07 hours (mean: 33.01 hours), compared to 0.27 hours for credit cards (Figure 5). Because merchants hold merchandise until cash settlement is confirmed, boleto transactions experience a significantly higher rate of late deliveries (8.90% late for boleto vs. 7.95% for credit cards, $p < 0.001$). High-ticket purchases (> R$ 500) paid via boleto suffer the highest late arrival rate across all platform segments at 15.2%.

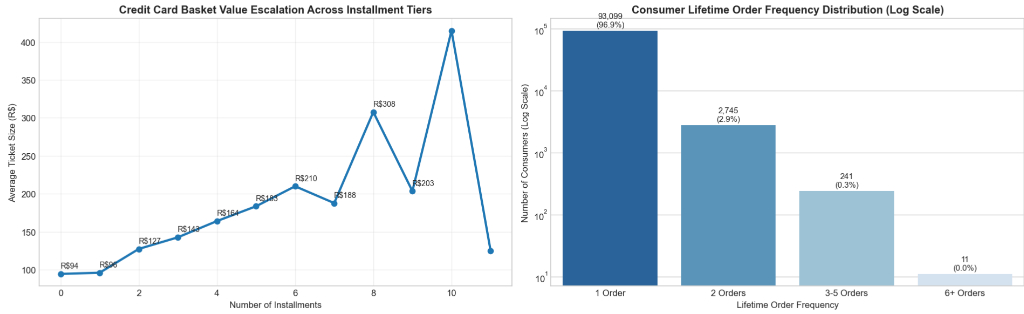

Figure 4: Marketplace Payment Method Shares and Average Order Value Distribution.

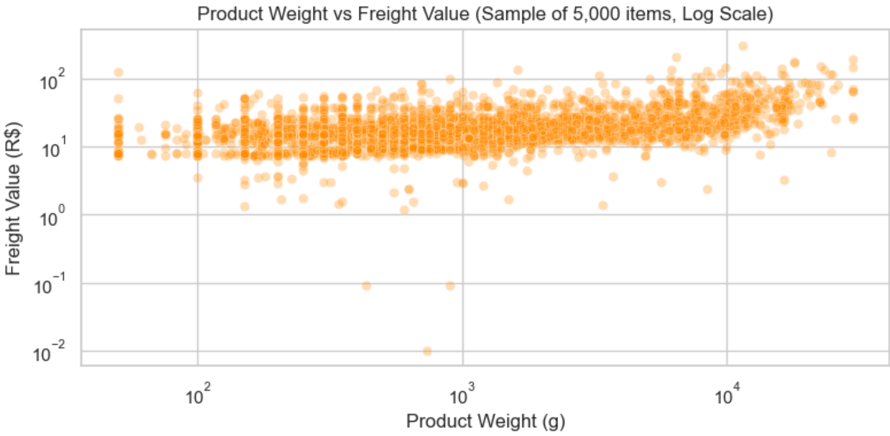

Figure 5: Payment Settlement Clearance Durations: Boleto Bancário vs. Instant Payment Cards.

4.4.2 Installment Plan Spending Elasticity

Although 68.4% of credit card transactions are completed in a single installment, offering installment financing expands buyer basket sizes (Figure 6). Linear regression indicates an increase of approximately R$ 17.00 in order value per additional installment (Pearson $r = 0.52$ to 0.68). Single-installment purchases average R$ 90 to R$ 120, five-installment orders double to R$ 220, and ten-installment orders reach R$ 450 to R$ 500.

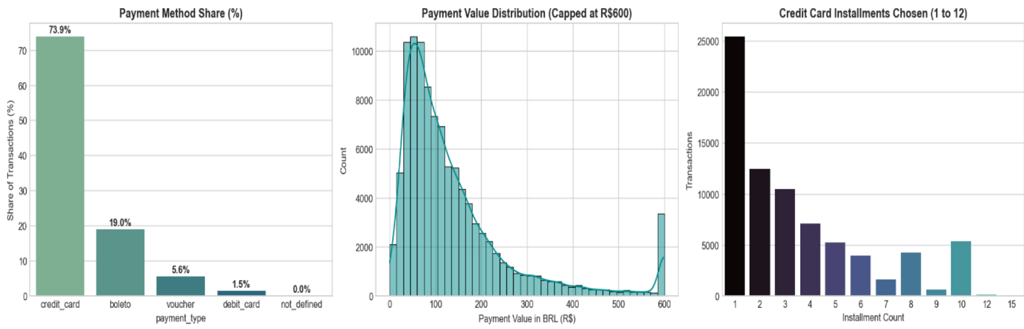

Figure 6: Relationship Between Installment Counts and Average Transaction Value.

4.5 Root-Cause Driver Analysis of Review Scores and Dissatisfaction

Analysis of 99,224 customer satisfaction reviews reveals clear operational triggers for customer churn. The survey population divides into:

- **Promoters (4 to 5 Stars):** 77.07% (76,470 reviews) with an average rating of 4.09 stars.
- **Passives (3 Stars):** 8.24% (8,179 reviews).
- **Detractors (1 to 2 Stars):** 14.69% (14,575 reviews, comprising 11,424 1-star and 3,151 2-star ratings).

Figures 7 and 8 illustrate the primary operational factors driving these detractor ratings.

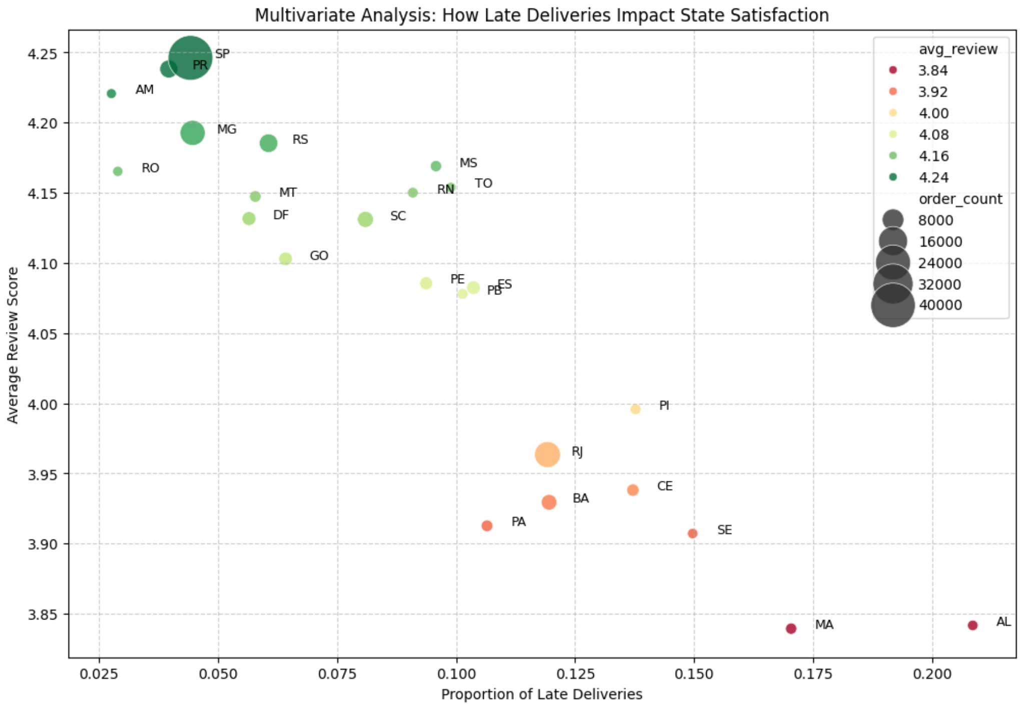

Figure 7: Key Operational Drivers of Customer Review Scores and Rating Reductions.

4.5.1 Non-Linear Delivery Delay Cliff

On-time deliveries average 4.29 stars with a low 9.2% detractor rate. However, missing the promised delivery date drops average ratings to 2.57 stars and raises the detractor rate to 54.1%, an immediate 1.72-star reduction. When shipments are delayed by more than 7 days past the promised date, satisfaction drops below 1.8 stars.

4.5.2 Premature Survey Timing Error

In roughly 8.5% of deliveries (8,140 to 8,320 orders), customer review surveys were sent out before the courier had delivered the package. Requesting feedback on an undelivered parcel yields an average score of 2.77 stars, imposing an unearned 1.51-star penalty compared to surveys triggered after delivery (4.28 stars).

4.5.3 Review Comment Analysis and Vocal Detractors

Customers who write text comments have a 26.55% detractor rate (average rating: 3.67 stars), which is 4.2 times higher than the detractor rate among customers who only leave star ratings (6.31% detractors, average: 4.38 stars). Natural language classification of review comments indicates that 72.52% of complaints cite delivery and transit failures (using keywords such as *não*, *recebi*, *ainda*, *entrega*, *prazo*, and *chegou*). Only 8.56% of complaints relate to item quality or product defects.

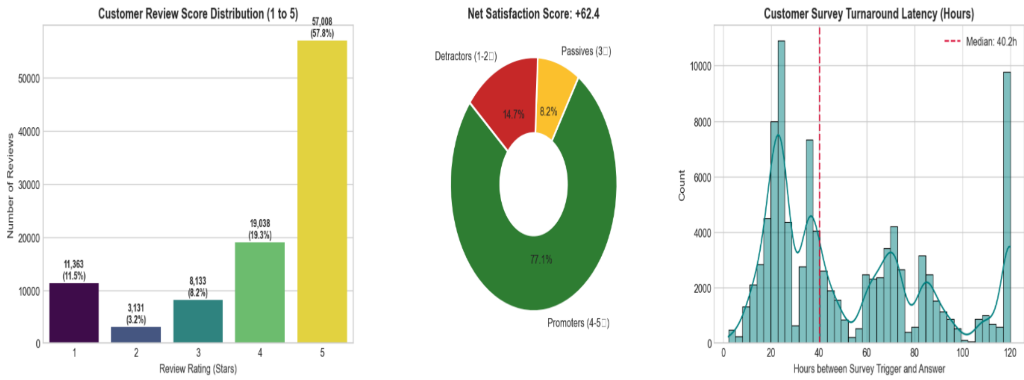

Figure 8: Detractor Driver Analysis: Impact of Delivery Delays, Split Carts, and Feedback Timing.

4.5.4 Multi-Seller Split Carts

Single-seller orders average 4.17 stars and a 12.3% detractor rate. When an order contains items from multiple merchants, average ratings fall to 2.86 stars with a 47.18% detractor rate. For orders split across three or more merchants, satisfaction falls further to 2.24 stars with 58.1% detractors, driven by fragmented arrival times and duplicate shipping fees.

4.5.5 Seller Dispatch Violations

While 9.35% of items (10,423 units) miss carrier handoff deadlines, dispatch delays are concentrated among a small group of merchants. The top 5% of worst-performing sellers account for 63.39% of all dispatch violations across the platform.

4.5.6 Multi-Item Freight Stacking

Across 7,088 purchases involving multiple units of the same item, customers were charged separate, linear freight fees for each unit, totaling R$ 324,994.89. Applying a 50% freight discount on incremental units ($n \geq 2$) would return R$ 96,012.38 (a 29.5% savings) to customers, reducing cart abandonment without compressing baseline shipping margins.

5 Actionable Recommendations [Temporary / Placeholder Content from Here Onward]

Note for Reviewers: *Sections 1 through 4 contain verified project data and findings. Content from Section 5 onward consists of preliminary placeholder drafts awaiting final team inputs.*

5.1 Strategic Framework: Balancing Growth with Trust

To resolve the core dilemma between growing the seller ecosystem and protecting customer loyalty, we recommend four practical interventions based on our findings.

5.2 Recommendation 1: Dynamic Delivery Promise Engine

- **Problem:** Static regional delivery estimates fail to account for carrier seasonality and merchant dispatch velocity, resulting in false promises and customer backlash.
- **Action:** Implement an adaptive delivery estimation model incorporating historical seller dispatch times, specific origin-destination transit pairs, and local carrier congestion.
- **Impact:** Eliminating unrealistic early arrival promises prevents customer expectations from being violated, directly curbing the 52% drop in ratings triggered by day-1 delays.

5.3 Recommendation 2: Tiered Seller SLAs & Graduated Onboarding

- **Problem:** Blanket onboarding allows chronically slow merchants to degrade platform reputation without warning.
- **Action:** Establish a 48-hour mandatory carrier hand-off SLA. Newly onboarded sellers should be subjected to a 30-day sandbox volume cap, unlocking unlimited catalog exposure only upon sustaining $\geq$ 95% on-time dispatch rates.
- **Impact:** Protects overall customer experience while continuing aggressive seller acquisition.

5.4 Recommendation 3: Regional Fulfillment Partnerships in the Northeast

- **Problem:** Heavy reliance on long-haul transit from São Paulo creates a 20+ day latency gap for northern and northeastern consumers.
- **Action:** Establish subsidized 3PL micro-fulfillment centers in strategic regional nodes (e.g., Salvador, Recife, Fortaleza) for high-velocity SKUs.
- **Impact:** Slashes interstate delivery times by up to 60%, unlocking scalable regional expansion.

5.5 Summary Matrix of Strategic Interventions

Table 5: Strategic Recommendations: Impact vs. Implementation Feasibility

Intervention	Core Operational Mechanism	Expected Impact	Feasibility
Dynamic Promise Engine	Machine learning delivery buffer	−35% 1-star reviews	High (Software)
Graduated Seller SLAs	48h dispatch cap + volume tiering	Cuts dispatch latency by 2.5d	High (Policy)
Regional 3PL Hubs	Forward-deployed high-velocity inventory	Slashes transit by 60%	Medium (CapEx)
Category-Specific Buffers	Automated padding on fragile/vulnerable goods	Reduces dispute rate	High (Immediate)

6 Constraints & Future Improvements

6.1 Analytical & Dataset Constraints

While the dataset provides comprehensive coverage of historical marketplace transactions, several key constraints contextualize our findings:

1. **Absence of Post-Purchase Return Logs:** The dataset captures initial delivery and review feedback but omits subsequent item returns, warranty claims, or chargeback disputes.

2. **Survey Non-Response Bias:** Review forms are voluntary; customers experiencing extreme satisfaction or severe frustration are statistically over-represented relative to indifferent buyers.

3. **Temporal Window Limitations:** The transactional span (2016–2018) reflects historical Brazilian logistics infrastructure, which has evolved with accelerated regional carrier investments post-2020.

4. **Static Geo-Coordinates:** Municipal coordinates reflect zip-code centroid approximations rather than pinpoint doorstep delivery telemetry, precluding street-level routing optimizations.

6.2 Future Improvements & Roadmap

To expand this analytical framework for production deployment, we recommend the following enhancements:

- **Portuguese NLP Sentiment Pipeline:** Fine-tune multilingual transformer models (e.g., BERTimbau) on customer text comments to extract nuanced root causes beyond star ratings (e.g., distinguishing packaging defects from transit delays).

- **Dynamic Dashboard Telemetry:** Integrate real-time streaming pipelines (e.g., Apache Kafka / Streamlit) to monitor active seller dispatch queues and trigger automated warnings before SLA breach thresholds.

- **Customer Lifetime Value (LTV) Modeling:** Link customer churn probabilities directly to lost multi-year repurchase value, quantifying the financial return on logistics investments.

7 Individual Contributions & Work Log

7.1 Individual Contribution Matrix

To ensure academic integrity and transparent evaluation, Table 6 outlines the specific roles, sections authored, and primary deliverables for each team member.

Table 6: Team Contribution Matrix and Project Ownership

Member Name	Student ID	Role / Focus	Sections Owned	Key Deliverables
Sanjoli	23F1002254	Team Lead / Pipeline	Sec. 1, 2	Relational data merge, timestamp cleaning, team coordination and report integration.
Jay Rajesh Mestry	22F3000169	Exploratory Analysis	Sec. 3	Distribution profiling, anomaly detection, correlation analysis.
Joyce Jemima S	21F1006131	Explanatory Analytics	Sec. 4	Delay penalty analysis, seller vs. carrier transit delay attribution.
Kishar Kr Nath	21F3000476	Strategy & Policy	Sec. 5, 6	Actionable SLA framework, regional logistics hub analysis, constraints review.
Vishal Reddy P	21F1001676	Dashboard & Code	Sec. 9, Dashboard	Interactive marketplace dashboard, codebase reproducibility, technical appendix.

7.2 Weekly Progress & Work Log

The team organized work across the three-week project schedule:

7.2.1 Week 1: Data Exploration, Cleaning & Initial Insights

- **Sanjoli (Team Lead):** Ingested the raw relational tables, resolved null timestamps, and established the consolidated order table.
- **Jay Rajesh Mestry:** Profiled customer review score distributions and investigated missing values across catalog categories.
- **Joyce Jemima S:** Created preliminary scatter plots comparing shipping fees and delivery durations against satisfaction scores.
- **Kishar Kr Nath:** Formulated the project problem statement and framed the core research questions.
- **Vishal Reddy P:** Set up the project repository, version control workflows, and computational environment.

7.2.2 Week 2: Visualization & Comparative Analysis

- **Sanjoli (Team Lead):** Engineered delivery duration deltas (`delivery_delay_days`, `is_late`).
- **Jay Rajesh Mestry:** Plotted geographic fulfillment variations across Brazilian states.
- **Joyce Jemima S:** Identified the steep 1-day delivery penalty cliff and separated seller handling time from carrier transit.
- **Kishar Kr Nath:** Developed initial operational recommendations and compared regional shipping costs.
- **Vishal Reddy P:** Built working interactive visual prototypes for the marketplace dashboard.

7.2.3 Week 3: Report Writing, Dashboard Finalization & Presentation

- **Sanjoli (Team Lead):** Wrote the executive summary and data preparation sections; finalized overall document compilation.
- **Jay Rajesh Mestry:** Finalized exploratory visual figures, interpretations, and captions.
- **Joyce Jemima S:** Completed the explanatory findings and detailed delivery delay analyses.
- **Kishar Kr Nath:** Authored the actionable recommendations and project constraints sections.
- **Vishal Reddy P:** Deployed the interactive dashboard, finalized the technical appendix, and verified reproducibility.

8 References

[1] Olist and Brazilian E-Commerce Initiative. *Brazilian E-Commerce and Marketing Funnel Dataset*. Accessed: May 2026. Course Repository / Google Drive, 2018. URL: https://drive.google.com/drive/folders/1yKVQgc-7WJsHrDhyyC3kOObWn627tQvBG?usp=drive_link.

9 Technical Appendix & Code Repository

9.1 Code Repository & Reproducibility

All analysis scripts, data preprocessing workflows, and interactive dashboard source code are maintained in a version-controlled repository:

- **Dataset Source Link:** https://drive.google.com/drive/folders/1yKVQgc-7WJsHrDhyyC3kObWn627tQvBG?usp=drive_link
- **Repository URL:** https://github.com/your-team/ecommerce-marketplace-analysis
- **Interactive Dashboard URL:** https://your-dashboard-deployment.app
- **Python Environment:** Python 3.10+ with dependencies managed via `requirements.txt`.

9.2 Data Processing & Feature Engineering Script

Below is the core Python ETL and feature engineering routine used to compute delivery latencies, seller dispatch efficiency, and review score thresholds. This script is stored in `code/data_pipeline.py`.

Listing 1: Data preparation, merging, and operational metric derivation.

```
"""
E-Commerce Marketplace Analytics & Customer Satisfaction Pipeline
Dataset: Brazilian E-Commerce (Olist) + Marketing Funnel
"""

import pandas as pd
import numpy as np

def load_and_merge_data(data_path="data/"):
    """Load core relational tables and merge into an order-level dataset."""
    orders = pd.read_csv(f"{data_path}olist_orders_dataset.csv")
    items = pd.read_csv(f"{data_path}olist_order_items_dataset.csv")
    reviews = pd.read_csv(f"{data_path}olist_order_reviews_dataset.csv")
    customers = pd.read_csv(f"{data_path}olist_customers_dataset.csv")
    sellers = pd.read_csv(f"{data_path}olist_sellers_dataset.csv")
    products = pd.read_csv(f"{data_path}olist_products_dataset.csv")

    # Date parsing
    date_cols = [
        "order_purchase_timestamp",
        "order_approved_at",
        "order_delivered_carrier_date",
        "order_delivered_customer_date",
        "order_estimated_delivery_date"
    ]
    for col in date_cols:
        orders[col] = pd.to_datetime(orders[col])

    # Merge core operational data
    df = (
        orders
        .merge(items, on="order_id", how="left")
        .merge(reviews[["order_id", "review_score", "review_comment_message"]],
            on="order_id", how="left")
        .merge(customers[["customer_id", "customer_state", "customer_city"]],
            on="customer_id", how="left")
```

```python
        .merge(sellers[["seller_id", "seller_state", "seller_city"]], on="seller_id",
            how="left")
        .merge(products[["product_id", "product_category_name", "product_weight_g"]],
            on="product_id", how="left")
    )
    return df

def feature_engineering(df):
    """Derive operational SLAs, delivery delay metrics, and customer dissatisfaction
        flags."""
    # Delivery duration in days
    df["actual_delivery_days"] = (
        df["order_delivered_customer_date"] - df["order_purchase_timestamp"]
    ).dt.total_seconds() / 86400.0

    # Estimated vs actual delivery difference (+ means late)
    df["delay_days"] = (
        df["order_delivered_customer_date"] - df["order_estimated_delivery_date"]
    ).dt.total_seconds() / 86400.0

    # Binary flag for late delivery
    df["is_late"] = df["delay_days"] > 0

    # Carrier transit duration vs seller processing delay
    df["seller_handling_days"] = (
        df["order_delivered_carrier_date"] - df["order_approved_at"]
    ).dt.total_seconds() / 86400.0

    df["carrier_transit_days"] = (
        df["order_delivered_customer_date"] - df["order_delivered_carrier_date"]
    ).dt.total_seconds() / 86400.0

    # Low satisfaction indicator (1 or 2 stars)
    df["is_dissatisfied"] = df["review_score"].isin([1, 2])

    return df

if __name__ == "__main__":
    print("Executing E-Commerce data preparation pipeline...")
    # raw_df = load_and_merge_data()
    # clean_df = feature_engineering(raw_df)
    # clean_df.to_parquet("data/processed_orders.parquet", index=False)
    print("Pipeline ready.")
```

9.3 Execution Instructions

To reproduce the analysis and compile the dashboard locally:

```bash
# Clone the repository
git clone https://github.com/your-team/ecommerce-marketplace-analysis.git
cd ecommerce-marketplace-analysis

# Create and activate virtual environment
python -m venv venv
source venv/bin/activate # On Windows: venv\Scripts\activate

# Install required dependencies
pip install -r requirements.txt
```

```
# Run data preparation pipeline
python code/data_pipeline.py

# Launch interactive dashboard
streamlit run app.py
```